

DBMS – SELLER AND INVENTORY MANAGEMENT

1. Database Design

1.1 Seller Table

Attribute	Data Type	Constraint
Seller_ID	INT	Primary Key
Seller_Name	VARCHAR(100)	NOT NULL
Email	VARCHAR(100)	UNIQUE
Phone	VARCHAR(15)	-
Address	VARCHAR(200)	-

1.2 Inventory Table

Attribute	Data Type	Constraint
Inventory_ID	INT	Primary Key
Product_ID	INT	Foreign Key
Stock_Quantity	INT	NOT NULL
Reorder_Level	INT	-
Last_Updated	DATE	-

2. Table Relationships

Product → Inventory

- Product.Product_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Product_ID → Foreign Key referencing Product.Product_ID
- **Relationship:** One-to-One (1:1)

3. Table Creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID INT PRIMARY KEY,  
3     Seller_Name VARCHAR(100) NOT NULL,  
4     Email VARCHAR(100) UNIQUE,  
5     Phone VARCHAR(15)  
6 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID INT PRIMARY KEY,  
3     Product_ID INT,  
4     Stock_Quantity INT NOT NULL,  
5     Reorder_Level INT,  
6     Last_Updated DATE,  
7     FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
8 );
```

Table created.

4. Seller Data Insertion

```
SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone)  
2 VALUES (1, 'ABC Electronics', 'abc@gmail.com', '9876543210');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone)  
2 VALUES (2, 'Tech World', 'techworld@gmail.com', '9876543211');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone)  
2 VALUES (3, 'Smart Retail', 'smartretail@gmail.com', '9876543212');  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone)  
2 VALUES (4, 'Fashion Hub', 'fashionhub@gmail.com', '9876543213');  
  
1 row created.
```

5. Inventory Data Insertion

```

SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone, Address) VALUES (1, 'ABC Electronics',
'abc@gmail.com', '9876543210', 'Chennai');

1 row created.

SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone, Address) VALUES (2, 'Tech World',
'techworld@gmail.com', '987654321', 'Coimbatore');

1 row created.

SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone, Address) VALUES (3, 'Smart Retail',
'smartretail@gmail.com', '9876543212', 'Madurai');

1 row created.

SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone, Address) VALUES (4, 'Fashion Hub',
'fashionhub@gmail.com', '9876543213', 'Trichy');

1 row created.

SQL> INSERT INTO Seller (Seller_ID, Seller_Name, Email, Phone, Address) VALUES (5, 'Home Store',
'homestore@gmail.com', '9876543214', 'Salem');

1 row created.

```

6. Seller Product Information

6.1 Display Seller and Product Details

```

SQL> SELECT
2     S.Seller_ID,
3     S.Seller_Name,
4     P.Product_ID,
5     P.Product_Name,
6     P.Brand,
7     P.Price,
8     P.Stock
9 FROM Seller S
10 JOIN Product P
11 ON S.Seller_ID = P.Seller_ID
12 ORDER BY S.Seller_ID;

```

SELLER_ID	SELLER_NAME	PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK
1					
2					
3					
4					
5					

7. Inventory Status

7.1 Display Available Products

```
SQL> SELECT
2     Product_ID,
3     Product_Name,
4     Brand,
5     Price,
6     Stock
7 FROM Product
8 WHERE Stock > 0;
```

PRODUCT_ID

PRODUCT_NAME

BRAND

	PRICE	STOCK
101		
iPhone 15		
Apple		
69999	25	

PRODUCT_ID

PRODUCT_NAME

BRAND

	PRICE	STOCK
102		
Samsung Galaxy S24		
Samsung		

7.2 Display Unavailable Products

```
SQL> SELECT
2     Product_ID,
3     Product_Name,
4     Brand,
5     Price,
6     Stock
7 FROM Product
8 WHERE Stock = 0;
```

no rows selected

8. Inventory Update

Update stock quantity and status.

```
SQL> UPDATE Product
  2 SET Stock = 50
  3 WHERE Product_ID = 101;

1 row updated.
```

9.Inventory status reports

9.1 Complete inventory Report

```
SQL> SELECT * FROM Inventory;
```

INVENTORY_ID	PRODUCT_ID	STOCK_QUANTITY	REORDER_LEVEL	LAST_UPDA
1	101	25	5	21-SEP-26
2	102	30	5	21-SEP-26
3	103	40	10	21-SEP-26
4	104	15	5	21-SEP-26
5	105	20	5	21-SEP-26
6	106	18	5	21-SEP-26
7	107	100	20	21-SEP-26
8	108	75	15	21-SEP-26
9	109	35	10	21-SEP-26
10	110	25	5	21-SEP-26
11	111	10	3	21-SEP-26
12	112	30	5	21-SEP-26

12 rows selected

9.2 Available and Unavailable Product Count

```
SQL> SELECT COUNT(*) AS Unavailable_Product_Count
  2 FROM Product
  3 WHERE Stock = 0;

UNAVAILABLE_PRODUCT_COUNT
-----
0
```

9.3 Total Stock By Seller

```

SQL> SELECT
  2     S.Seller_ID,
  3     S.Seller_Name,
  4     SUM(P.Stock) AS Total_Stock
  5 FROM Seller S
  6 JOIN Product P
  7     ON S.Seller_ID = P.Seller_ID
  8 GROUP BY S.Seller_ID, S.Seller_Name
  9 ORDER BY S.Seller_ID;

```

SELLER_ID

SELLER_NAME

TOTAL_STOCK

1	ABC Electronics	65
---	-----------------	----

2	Tech World	85
---	------------	----

SELLER_ID

SELLER_NAME

TOTAL_STOCK

3	Smart Retail	83
---	--------------	----

9.4 Products with Zero Stock

Display products that are currently out of stock.

```

SQL> SELECT COUNT(*) AS Zero_Stock_Product_Count
  2 FROM Product
  3 WHERE Stock = 0;

```

ZERO_STOCK_PRODUCT_COUNT

0